Status	Finished
Started	Wednesday, 21 October 2026, 4:13 PM
Completed	Wednesday, 21 October 2026, 4:17 PM
Duration	3 mins 11 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

- a. They are compared element by element
- b. They are combined to form a single sorted list
- c. They are split again into smaller sublists
- d. They are discarded

The correct answer is: They are combined to form a single sorted list

Question **2**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after dividing the input into smaller parts?

- a. The parts are ignored
- b. The parts are merged without sorting
- c. Each part is sorted independently
- d. Each part is searched for a specific element

The correct answer is: Each part is sorted independently

Question **3**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

- a. 40,70,80,90,120,150,200,300,400,470
- b. 200,470,80,150,40,90,300,400,70,120
- c. 80,150,200,470,40,90,300,400,70,120
- d. 40,80,90,150,200,300,400,470,70,120

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **4**

Complete

Mark 1.00 out of 1.00

The process of placing or rearranging a collection of elements into a particular order is known as

- a. Searching
- b. Merging
- c. Rearranging
- d. Sorting

The correct answer is: Sorting

Question **5**

Complete

Mark 1.00 out of 1.00

What does the Bubble Sort algorithm primarily focus on during each pass?

- a. Bubbling up the largest element to its correct position
- b. Sorting the entire list in one pass
- c. Dividing the list into halves
- d. Bubbling up the smallest element

The correct answer is: Bubbling up the largest element to its correct position

Question **6**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

- a. Makes data harder to manage
- b. Decreases the efficiency of algorithms
- c. Provides a basis for other algorithms to work efficiently
- d. Makes code execution slower

The correct answer is: Provides a basis for other algorithms to work efficiently

Question **7**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

- a. sorted()
- b. arrange()
- c. order()
- d. sort()

The correct answer is: sorted()

Question **8**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

a. It has no effect on the analysis process
b. It complicates the analysis process
c. It allows for quicker identification of duplicates
d. It makes the analysis slower

The correct answer is: It allows for quicker identification of duplicates

Question **9**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

a. Sorting the entire list sequentially
b. Comparing each element with the others
c. Dividing the input into smaller subproblems
d. Combining sorted sublists

The correct answer is: Dividing the input into smaller subproblems

Question **10**

Complete

Mark 1.00 out of 1.00

What is a key disadvantage of Bubble Sort compared to more advanced algorithms like Merge Sort?

a. Bubble Sort does not guarantee sorted order
b. Bubble Sort cannot handle duplicate elements
c. Bubble Sort is less efficient for large lists
d. Bubble Sort is difficult to implement

The correct answer is: Bubble Sort is less efficient for large lists

Question **11**

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

a. sorted()
b. sort()
c. arrange()
d. order()

The correct answer is: sort()

Question **12**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

a. Linear Search
b. Binary Search
c. Merge Sort
d. Bubble Sort

The correct answer is: Bubble Sort

Question **13**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the term "sorting" in computer science?

a. Arranging data in a specific order
b. Finding a specific element in a list
c. Removing duplicates from a list
d. Merging two datasets

The correct answer is: Arranging data in a specific order

Question **14**

Complete

Mark 1.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

a. Quick Sort
b. Merge Sort
c. Bubble Sort
d. Linear Search

The correct answer is: Merge Sort

Question **15**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

a. Merge Sort
b. Binary Search
c. Linear Search
d. Bubble Sort

The correct answer is: Merge Sort